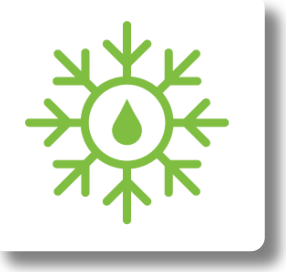
COOLING:
COLD PLATE



# Deep Dive: Biomanagement for Liquid Cooling


Dakota Bunn *Lead Microbiologist, Nalco Water an Ecolab Company*

Sean Sivapalan *Principal Engineer, NVIDIA*



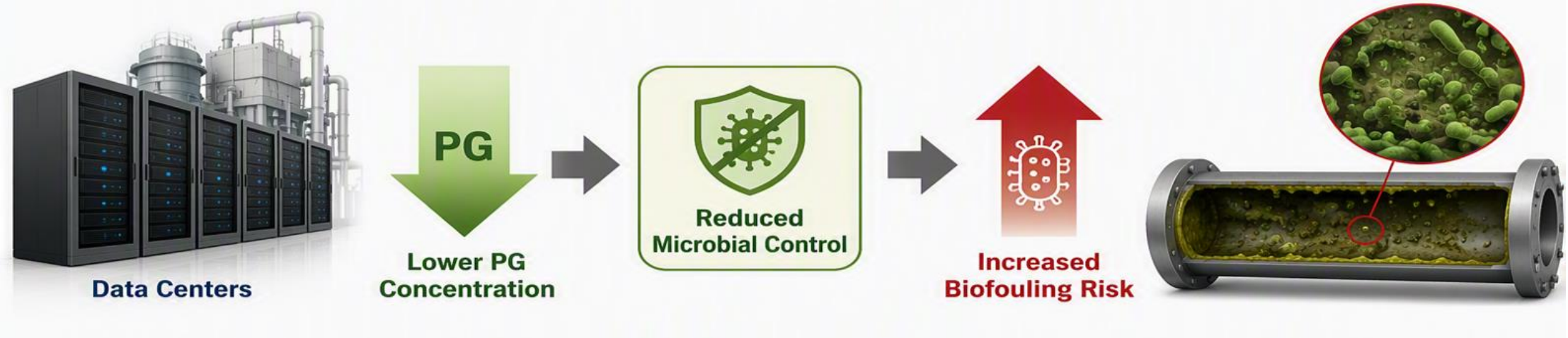

# Today’s Roadmap

**From contamination risk to practical takeaways**

1. **Microbial Contamination in Glycol Cooling Loops**
2. **What ≥25% Propylene Glycol Really Means**
3. **Control Strategies**
4. **Microbial Monitoring**
5. **Risk-Based Monitoring Framework**
6. **Key Takeaways and Call to Action**

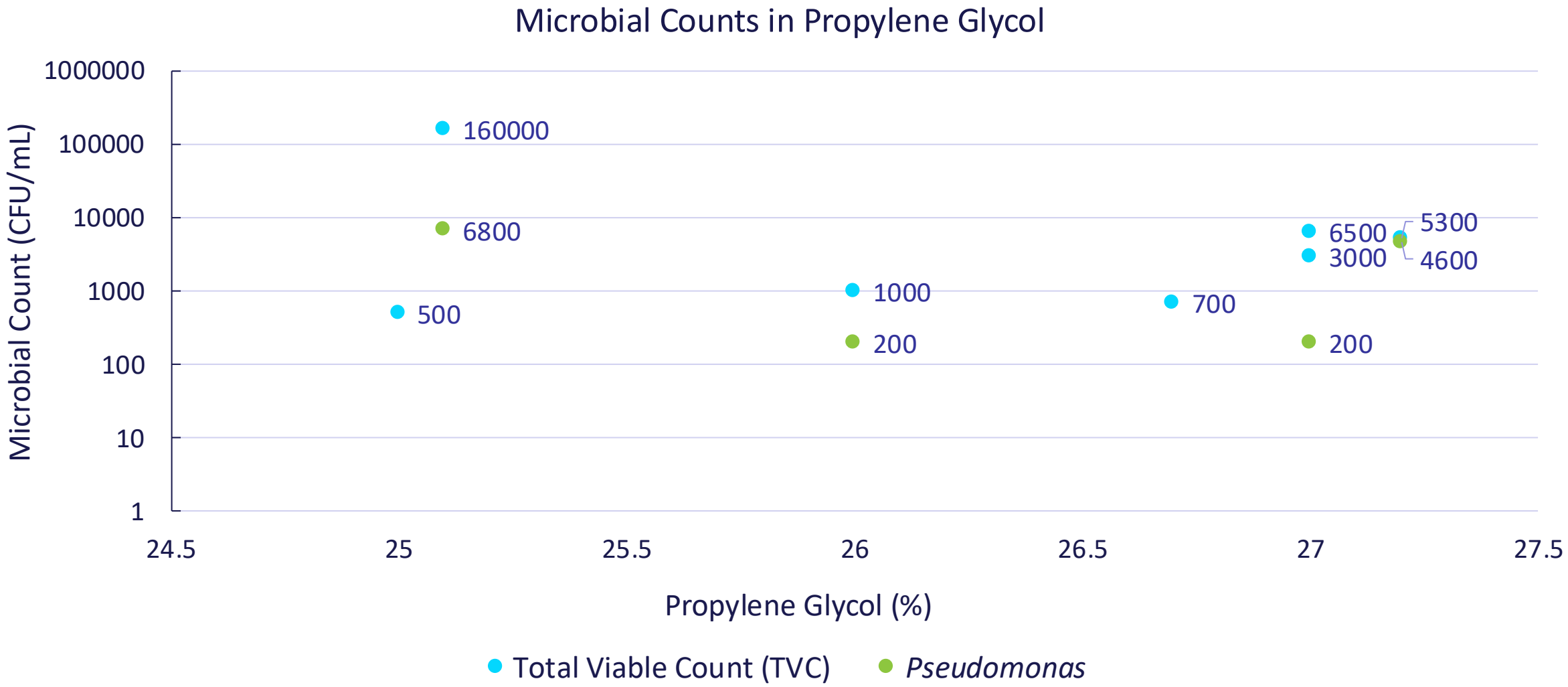
# Contamination in Glycol Cooling Loops

**Desire to lower propylene glycol concentration to improve thermal performance**




- Major Concerns: polymicrobial populations, biofilm formation, sampling challenges
- Common Entry: Make-up/dilution water, maintenance events, and air ingress
- ≥25% propylene glycol decreases unplanned downtime while limiting bacterial risk

# Real World Contamination Examples



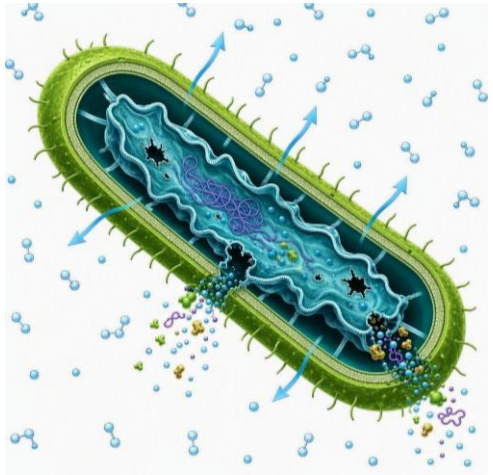
# What Does ≥ 25% PG Really Mean?

**At ≥ 25% PG, biological contaminants are dormant**

↓

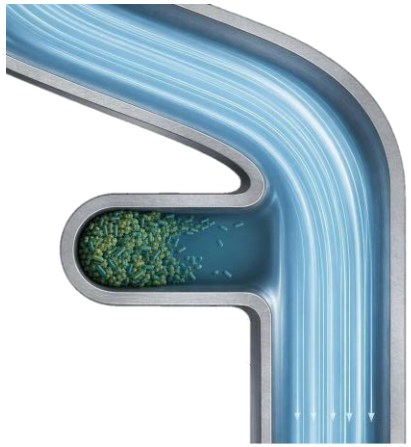
**BioSTATIC**

**Mechanism of Action**



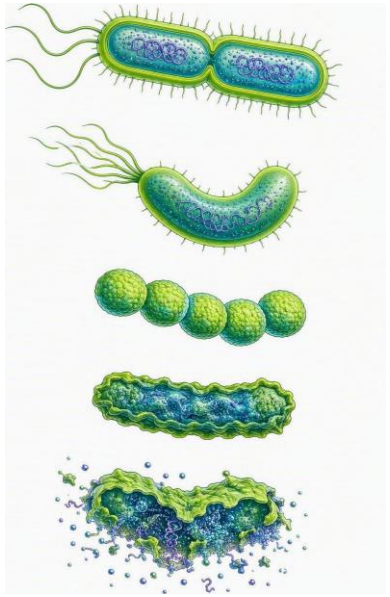
↑ **Osmotic pressure**
**Dehydrates cell**
**Disrupts cell structures**

**Viable organisms persist at surfaces**



**And in low-flow niches such as dead legs**

**Tolerances to PG concentration vary by bacteria**

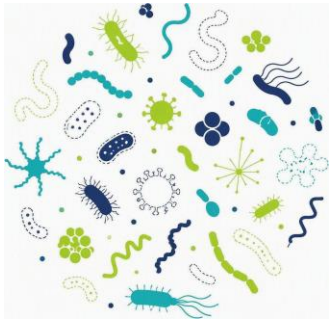


**Below ≥ 25% PG > osmotic pressure decreases > growth and biofilm formation can resume**

# Why <25% Isn’t a Guarantee

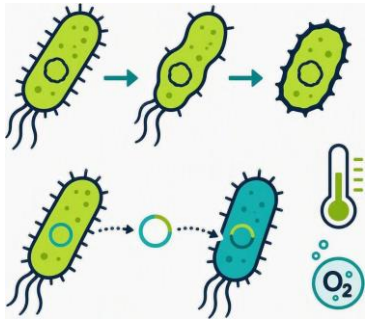
**System complexity governs microbial control, not a single concentration limit**

### Microbial diversity



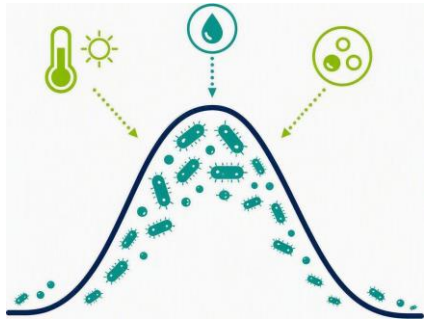
Environmental communities are highly varied and largely unculturable

### Microbial adaptability




Extremophiles, flexible respiration, and gene transfer enable survival

### Population dynamics

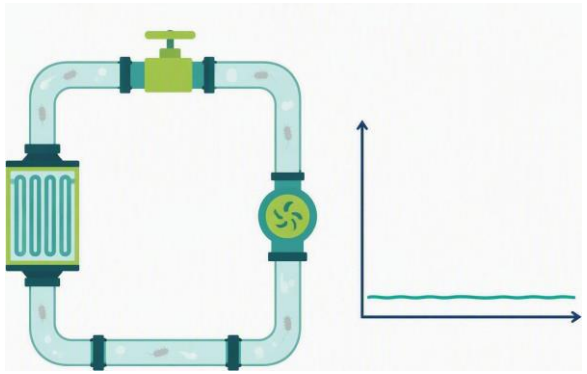


Temperature, nutrients, oxygen, and community shifts move the effective threshold

**Best understood as a practical engineering guideline, not a definitive boundary**

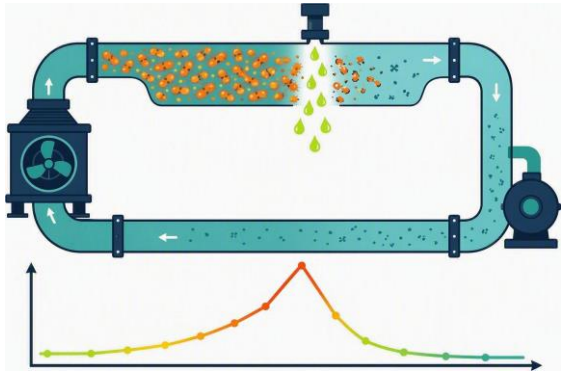
# Biostatic vs. Biocidal Control

## Biostatic: ≥25% PG baseline



- Passive, simple chemistry (no biocide handling)
- Holds low, stable counts (~$10^2$-$10^3$ CFU/mL)
- Cannot kill or “fight back” if control slips

## Biocidal: reactive remediation



- Active kill that remediates contamination
- Adds chemistry and compatibility concerns
- May be needed when heavy bio-challenge requires system cleaning and remediation

**Many fluid providers do not recommend biocides: always consult your provider before dosing (warranty risk)**

# Biocide Categories and Mechanisms

**Prefer to avoid: know the options if remediation becomes necessary**

## Oxidizing
Chlorine, bromine, hydrogen peroxide



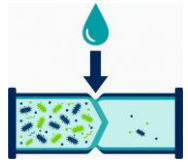
Fast: but attacks glycol, inhibitors, and raises corrosion risk

## Non-oxidizing (Slow)
Isothiazolin, glutaraldehyde, bronopol



Slower, usually longer-term compatibility

## Non-oxidizing (Fast)
DBNPA



Rapid knockdown of active contamination but lacks long term compatibility



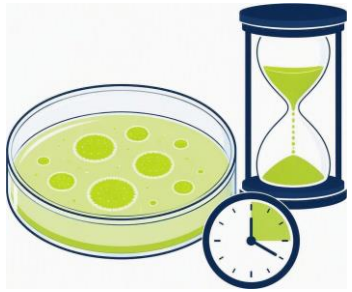
**If you do dose:**

- Select biocides proactively at start-up, per the fluid provider's specification
- Confirm compatibility at use concentration: high pH (~8-10), glycol, inhibitors, and every wetted material (mixed metals, elastomers, etc.)

# Monitoring: Colony Counts vs Metabolic Testing

## Plate counts (CFU/mL)

Counts cells that will grow on agar



- Answer in 24 to 48+ hours
- Bacterial type specific (some organisms never grow)
- Can miss viable but non-culturable cells

## ATP metabolic testing (RLU)

Measures how much life is active right now



- Answer in minutes, at the sample point
- Flags active growth before it spreads
- Total ATP - free ATP = real activity



**The biostatic difference: PG holds organisms in check rather than killing them**

- Counts can stay high while ATP stays low: the population is present but not proliferating
- Trend both together; a single reading can be misleading

# A Proposed Risk-Based Monitoring Framework

## Match sampling frequency to contamination potential

| Risk Level | What It Looks Like | EXAMPLE Sampling Frequency* |
|---|---|---|
| **Low** | Stable, isolated, tight control | **Bi-annual verification sampling** |
| **Medium** | Periodic service, node change-outs | **Quarterly testing** |
| **High** | Frequent opening, high dilution potential, history of air ingress | **Monthly or event-triggered** |

**Goal: detect loss of control early: before biofilm, performance loss, or corrosion**

*Recommended frequency to be determined by OCP fluids workstream*

# Key Takeaways & Call to Action

1. Maintain ≥ 25% PG as the primary, passive biostatic defense
   - Treat it as a guideline, not a guarantee
2. Control the primary triggers for microbial rebound:
   - Dilution and fill-water quality, node change outs, design to limit air ingress
3. Apply a risk-based sampling cadence, consult your fluid provider before any biocide remediation program
4. Pair plate counts and metabolic tests and interpret trends, not single data points

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Get involved and shape OCP biomanagement guidance for liquid-cooled data centers

- Learn more & contribute: https://www.opencompute.org

# Thank You!

